

ABUBAKER ATTIQUE

AI/ML Engineer | M.Sc. Artificial Intelligence Student

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Professional Summary

M.Sc. Artificial Intelligence student at BTU Cottbus-Senftenberg with a B.Sc. in Computer Science and 2+ years of professional software development experience. Designs, trains, and deploys deep learning and full-stack applications across computer vision, speech, NLP, and multi-modal AI. Hands-on with LLM and RAG systems, vector databases, and production deployment using Python, PyTorch, TensorFlow, FastAPI, Docker, and ReactJS. Seeking AI/ML Engineer, Data Science, or Working Student (Werkstudent) roles in Germany.

Professional Experience

Developer

R2V Private Limited

May 2025 – May 2026

Islamabad, Pakistan

- Led frontend development and feature expansion for web and GIS applications, owning delivery from requirements to production deployment.
- Refactored legacy React code into reusable modular components, improving maintainability and reducing rendering bottlenecks.
- Built interactive dashboards, data filters, and GIS map tools with backend integration, optimizing load times and UI stability.
- Acted as technical coordinator, translating stakeholder requirements into production-ready features and supporting testing and deployment.

React Front-End Developer

R2V Private Limited

Oct 2024 – Mar 2025

Islamabad, Pakistan

- Developed responsive, cross-device interfaces in ReactJS with reusable components and robust form validation.
- Partnered with designers to improve UI/UX and frontend performance; resolved UI defects through systematic debugging.

AI Engineer (Internship)

LIMS-COE

Jun 2024 – Sep 2024

Rawalpindi, Pakistan

- Trained a CNN-based deep learning model for plant-disease classification, aggregating and standardizing datasets from diverse sources to ensure data integrity.
- Integrated the trained model into the organization's production application, enabling real-time diagnostics.
- Contributed research and insights to technical working papers and presentations for the engineering team.

React Front-End Developer (Internship)

R2V Private Limited

Jun 2023 – Aug 2023

Islamabad, Pakistan

- Designed and built the frontend of the Balochistan Land Record MIS (BLRMIS) using React, Tailwind CSS, and Leaflet for interactive mapping.
- Collaborated with a cross-functional team of 4 developers to deliver frontend modules 2 weeks ahead of schedule.

Education

M.Sc. in Artificial Intelligence

Brandenburg University of Technology Cottbus-Senftenberg (BTU)

Apr 2026 – Present

Cottbus, Germany

B.Sc. in Computer Science

FAST-NUCES (National University of Computer & Emerging Sciences)

Sep 2020 – Jun 2024

Pakistan

Relevant Coursework: Artificial Intelligence, Applied Machine Learning, Natural Language Processing, Data Structures & Algorithms, Database Systems, Operating Systems, Software Engineering

Key Projects

FinMind AI – Privacy-First Financial Document Q&A (RAG)

2026

Python 3.12, FastAPI, Qdrant, Ollama, Pydantic, Docker Compose, ReactJS, github.com/AbuBakerAttique/FinMind-AI
Vite, pypdf

- Built a fully local Retrieval-Augmented Generation (RAG) platform that answers natural-language questions over financial PDFs with verifiable page-level citations, keeping sensitive data on-premise.
- Engineered the ingestion pipeline end to end: pypdf extraction, metadata-aware chunking, 768-dim embeddings, and Qdrant vector search with per-document filtering to prevent cross-document contamination.
- Eliminated LLM arithmetic errors with a deterministic calculator that validates extracted values via Pydantic and computes growth in Python Decimal, citing formula and source page for every figure.
- Shipped as a FastAPI service with a ReactJS dashboard, deployed via Docker Compose and covered by automated API, unit, and integration tests.

AI Interview Coach – Real-Time Multi-Modal Analysis

2026

Python, PyTorch, MediaPipe, OpenCV, Librosa, Flask, ReactJS github.com/AbuBakerAttique/AI-Interview-Coach

- Engineered a multi-modal AI pipeline with 4 custom deep learning models for real-time interview analysis via webcam and microphone.
- Trained from scratch a Speech CNN+LSTM on RAVDESS (2,880 audio files, MFCC features) reaching 93.4% accuracy, a Facial-Expression CNN on FER2013 (28,821 images) reaching 75.5%, and a Posture NN on MediaPipe landmarks (4,794 samples) reaching 97.3%.
- Built a score-fusion layer combining all models into a final confidence score with grade and actionable feedback.

AI Gym Rep Counter & Form Analyzer

2026

Python, MediaPipe, NumPy, Three.js, Web Speech API github.com/AbuBakerAttique/Gym-Project

- Built a real-time computer-vision fitness tracker using MediaPipe Pose to count reps and detect form errors across 6 exercises, with no paid APIs.
- Engineered a NumPy joint-angle engine and a state-machine rep counter with exercise-specific elbow, knee, and hip thresholds.
- Implemented phone-as-camera mode via MediaPipe.js over self-signed HTTPS, a Three.js 3D form guide, and real-time voice feedback.

Fake News Detector – Explainable AI Web Application

2026

Python, Flask, Streamlit, scikit-learn, NLTK, LIME fake-news-demo.streamlit.app

- Built a web app that classifies news articles as Fake or Real using TF-IDF features and machine learning models.
- Added LIME explanations, model comparison, word clouds, and statistical writing-style analysis (sentence structure, vocabulary, formality).

Mute Motion – AI Sign-Language to Text (Final-Year Project)

2023 – 2024

React Native, Firebase, Computer Vision

- Built an AI-driven sign-language recognition system enabling communication between sign-language and non-sign-language users, with a React Native frontend and Firebase backend for real-time interaction.

Technical Skills

Programming: Python, C/C++, SQL, JavaScript, PHP, HTML/CSS

AI/ML: PyTorch, TensorFlow, Keras, scikit-learn, CNNs, LSTMs, Computer Vision, OpenCV, MediaPipe, Librosa, LLMs, Retrieval-Augmented Generation (RAG), Vector Databases (Qdrant), Ollama, Prompt Engineering, Model Training, Fine-Tuning & Evaluation

Data Science: NumPy, Pandas, Matplotlib, Seaborn, Jupyter, Statistical Analysis, Data Visualization, Feature Engineering

Web & Backend: FastAPI, Flask, REST APIs, ReactJS, Next.js, React Native, Express.js, Tailwind CSS, Vite

Databases: PostgreSQL, MySQL, SQLite, MongoDB

Tools & DevOps: Git, GitHub, Docker, Docker Compose, Linux, VS Code, Figma, Web Scraping (BeautifulSoup)

Languages

English (Professional) | Urdu (Native) | Pashto (Native) | German (A1.2)